

# Final Project: Naive Bayesian Classifier and Bagging

**Introduction** This project evaluates a Naive Bayesian classifier and bagging ensembles on Banknote Authentication, Glass Identification, and Image Segmentation from the UCI Machine Learning Repository.

**Methodology** Continuous attributes are discretized by equal-width binning into ten bins. The discretizer is fit only on each training fold. Missing values are ignored for conditional probability estimation and skipped during prediction. Laplace estimates are used for the Naive Bayes probabilities. Bagging trains each base model on a bootstrap sample and combines predictions with majority vote. Performance is measured by five-fold cross validation using Chapter 4 method III: random numbers, sorting, and cyclic fold assignment.

| Results Dataset               | Model | Base              | Accuracy                |
|-------------------------------|-------|-------------------|-------------------------|
| ----- Banknote Authentication |       |                   |                         |
| NBC                           | 1     | 89.80% +/- 3.21%  | Banknote Authentication |
|                               |       | Bagging NBC (10)  | 10                      |
|                               |       | 89.65% +/- 3.67%  | Banknote Authentication |
|                               |       | Bagging NBC (20)  | 20                      |
|                               |       | 89.94% +/- 3.67%  | Banknote Authentication |
|                               |       | Bagging NBC (30)  | 30                      |
|                               |       | 90.09% +/- 3.44%  | Glass Identification    |
|                               |       | Bagging NBC (10)  | 10                      |
|                               |       | 61.73% +/- 12.46% | Glass Identification    |
|                               |       | Bagging NBC (20)  | 20                      |
|                               |       | 62.60% +/- 7.07%  | Glass Identification    |
|                               |       | Bagging NBC (30)  | 30                      |
|                               |       | 64.52% +/- 11.34% | Image Segmentation      |
|                               |       | NBC               | 1                       |
|                               |       | 80.00% +/- 5.98%  | Image Segmentation      |
|                               |       | Bagging NBC (10)  | 10                      |
|                               |       | 79.05% +/- 6.39%  | Image Segmentation      |
|                               |       | Bagging NBC (20)  | 20                      |
|                               |       | 80.00% +/- 6.43%  | Image Segmentation      |
|                               |       | Bagging NBC (30)  | 30                      |
|                               |       | 80.00% +/- 4.33%  |                         |

# Chapter 5 Single-Dataset Evaluation

Single Dataset Evaluation Chapter 5 matched-sample tests compare fold-by-fold differences on the same folds. Independent-sample tests are also shown as secondary comparisons.

| Dataset            | Comparison         | Diff.              | Match t              | Match p              | Indep t          | Indep p              |                      |
|--------------------|--------------------|--------------------|----------------------|----------------------|------------------|----------------------|----------------------|
| ----- Banknote     |                    |                    |                      |                      |                  |                      |                      |
| Authenticat        | Bagging NBC (10)   | -0.15%             | -0.2812              | 0.7925               | -0.0675          | 0.9479               | Banknote Authenticat |
| Bagging NBC (20)   | 0.15%              | 0.3866             | 0.7187               | 0.0666               | 0.9485           | Banknote Authenticat | Bagging NBC          |
| (30)               | 0.29%              | 0.5577             | 0.6068               | 0.1382               | 0.8935           | Glass Identification | Bagging NBC (10)     |
| 0.4002             | 0.7095             | 0.2040             | 0.8437               | Glass Identification | Bagging NBC (20) | 2.31%                | 1.0463               |
| 0.3545             | 0.4317             | 0.6784             | Glass Identification | Bagging NBC (30)     | 4.23%            | 1.2097               | 0.2930               |
| 0.6345             | 0.5439             | Image Segmentation | Bagging NBC (10)     | -0.95%               | -0.7845          | 0.4766               | -0.2434              |
| 0.8138             | Image Segmentation | Bagging NBC (20)   | 0.00%                | 0.0000               | 1.0000           | 0.0000               | 1.0000               |
| Image Segmentation | Bagging NBC (30)   | 0.00%              | 0.0000               | 1.0000               | 0.0000           | 1.0000               |                      |

| Single Dataset       | Bagging Size  | Comparison    | Dataset | Baseline | Comparison | Diff.         |
|----------------------|---------------|---------------|---------|----------|------------|---------------|
| Match t              | Match p       | Indep t       | Indep p |          |            |               |
| -----                |               |               |         |          |            |               |
| Banknote Authenticat | Bagging NBC ( | Bagging NBC ( | 0.29%   | 1.0918   | 0.3363     | 0.1259 0.9029 |
| Banknote Authenticat | Bagging NBC ( | Bagging NBC ( | 0.44%   | 1.6352   | 0.1773     | 0.1946 0.8506 |
| Banknote Authenticat | Bagging NBC ( | Bagging NBC ( | 0.15%   | 0.5874   | 0.5885     | 0.0646 0.9501 |
| Glass Identification | Bagging NBC ( | Bagging NBC ( | 0.87%   | 0.2200   | 0.8367     | 0.1366 0.8956 |
| Glass Identification | Bagging NBC ( | Bagging NBC ( | 2.79%   | 3.2071   | 0.0327     | 0.3705 0.7207 |
| Glass Identification | Bagging NBC ( | Bagging NBC ( | 1.92%   | 0.5566   | 0.6075     | 0.3207 0.7582 |
| Image Segmentation   | Bagging NBC ( | Bagging NBC ( | 0.95%   | 1.6330   | 0.1778     | 0.2349 0.8202 |
| Image Segmentation   | Bagging NBC ( | Bagging NBC ( | 0.95%   | 0.3885   | 0.7174     | 0.2760 0.7905 |
| Image Segmentation   | Bagging NBC ( | Bagging NBC ( | 0.00%   | 0.0000   | 1.0000     | 0.0000 1.0000 |

# Chapter 5 Multiple-Dataset Evaluation

Multiple Dataset Evaluation Across all three datasets, Bagging-30 has the largest average improvement over NBC, but neither the matched-sample nor independent-sample Chapter 5 test is significant at  $\alpha = 0.05$ .

| Baseline         | Comparison       |                  | Diff.            | Match t          | Match p | Indep t          | Indep p          |
|------------------|------------------|------------------|------------------|------------------|---------|------------------|------------------|
| ----- NBC        |                  |                  |                  |                  |         |                  |                  |
| Bagging NBC (10) | 0.11%            | 0.0888           | 0.9327           | 0.0407           | 0.9687  | NBC              | Bagging NBC (20) |
| 0.82%            | 0.9478           | 0.3798           | 0.3517           | 0.7341           | NBC     | Bagging NBC (30) | 1.51% 1.1336     |
| 0.3002           | 0.5848           | 0.5800           | Bagging NBC (10) | Bagging NBC (20) | 0.71%   | 0.5261           | 0.6266 0.2674    |
| 0.7969           | Bagging NBC (10) | Bagging NBC (30) | 1.39%            | 1.5987           | 0.1708  | 0.4870           | 0.6435 Bagging   |
| NBC (20)         | Bagging NBC (30) | 0.69%            | 0.4694           | 0.6530           | 0.2837  | 0.7848           |                  |

Conclusion Bagging improves mean accuracy for Banknote Authentication and Glass Identification and ties NBC on Image Segmentation for 20 and 30 base models. The improvements are positive but not statistically confirmed at the 0.05 level.